DERIVED HYPERCOVERING DESCENT: E -> RA_*A^-1E IS AN ISOMORPHISM IN THE DERIVED CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the author's 'urge the reader to skip' sentence is his own framing of the convergence technicality, not an omission - the double-complex argument that follows is complete and retained verbatim; the degree-zero isomorphism for injective sheaves cites the sheaf-level descent lemma delivered as card L46146.

Editorial note. The label lemma-hypercovering-descent-bounded-abelian names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Augmentation spectral sequence for the total site. Same chapter, \label{lemma-augmentation-spectral-sequence}. Line 4393: the spectral sequence whose degeneration on injectives opens the proof.
- a_p^-1 preserves injectives. Same chapter, \label{lemma-localize-injective}. Line 4396: (a^-1I)_p = a_p^-1I is injective, degenerating the spectral sequence; reused at line 4539 for the K-injective complex.
- Ra_*a^-1 computed by the hypercovering Cech complex. Same chapter, \label{lemma-hypercovering-cech-complex}. Line 4403: identifies the degenerated complex with SheafHom(s(Z_{F(K)}^#), I).
- s(Z_{F(K)}^#) is exact in positive degrees for a hypercovering. Hypercoverings chapter, \label{hypercovering-lemma-acyclic-hypercover-sheaves}. Lines 4407-4410: acyclicity of the simplicial presheaf underlying K; applied again at line 4488 to the quasi-isomorphism s(Z^#) -> Z.
- Sheaf-level hypercovering descent. Same chapter, \label{lemma-hypercovering-descent-sheaves} (own card, statement line 4166). Line 4424: I = a_*a^-1I gives the degree-zero isomorphism for injective sheaves.
- Leray's acyclicity lemma. Derived Categories chapter, \label{derived-lemma-leray-acyclicity}. Line 4431: derives the bounded-below case from the acyclicity of injectives.
- Simplicial resolution of the sheaf Z_U^# under a^-1. Same chapter, \label{lemma-simplicial-resolution-augmentation}. Lines 4460-4466: the quasi-isomorphism from the g_{p!}-complex to a^-1Z_U^# opening the unbounded case.

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- RSheafHom against a K-injective complex computes as a product total complex. Cohomology on Sites chapter, Section \ref{sites-cohomology-section-internal-hom}, \label{sites-cohomology-lemma-RHom-into-K-injective}, \label{sites-cohomology-lemma-section-RHom-over-U}. Lines 4472-4483 and 4505-4513: both RGamma sides become Tot_pi of Hom-term double complexes.
- j_! tensor identity for localized sheaves. Modules on Sites chapter, Remark \ref{sites-modules-remark-j-shriek-tensor}. Line 4527: Z^#_{K_p} tensor Z_U^# = a_{p!}a_p^-1Z_U^#, matching the terms of the two double complexes.
- g_p^-1 preserves K-injectivity. Same chapter, \label{lemma-restriction-injective-to-component-site}. Line 4541: g_p^-1J^bullet is K-injective, so the row complexes both compute RHom(a_p^-1Z_U^#, a_p^-1E).
- Product of termwise quasi-isomorphisms of double complexes. More on Algebra chapter, \label{more-algebra-lemma-prod-qis-gives-qis}. Line 4550: promotes the termwise quasi-isomorphism A^{p, bullet} -> B^{p, bullet} to the product-total quasi-isomorphism.

Source reference. Original lemma 015E: sources/stacks-chow/tags.txt, source lines 1505–1505. Prerequisite pointer invoked at proof line(s) 4429; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01GE: sources/stacks-chow/tags.txt, source lines 1903–1903. Prerequisite pointer invoked at proof line(s) 4414; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D7A: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 2020–2020. Prerequisite pointer invoked at proof line(s) 4398; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D8F: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 4354–4354. Prerequisite pointer invoked at proof line(s) 4409; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D8E: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 4167–4167. Prerequisite pointer invoked at proof line(s) 4422; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D88: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 3775–3775. Prerequisite pointer invoked at proof line(s) 4401; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WH: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1086–1086. Prerequisite pointer invoked at proof

line(s) 4539; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D78: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1929–1929. Prerequisite pointer invoked at proof line(s) 4459; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0EYX: sources/stacks-chow/tags.txt, source lines 18707–18707. Prerequisite pointer invoked at proof line(s) 4553; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0A94: sources/stacks-chow/tags.txt, source lines 12937–12937. Prerequisite pointer invoked at proof line(s) 4490; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08JA: sources/stacks-chow/tags.txt, source lines 10843–10843. Prerequisite pointer invoked at proof line(s) 4491; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 08J7: sources/stacks-chow/tags.txt, source lines 10840–10840. Prerequisite pointer invoked at proof line(s) 4480; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 0EYY: sources/stacks-chow/tags.txt, source lines 18708–18708. Prerequisite pointer invoked at proof line(s) 4528; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{C}$ be a site with equalizers and fibre products. Let K be a hypercovering. For $E \in D(\mathcal{C})$ the map*

$$E \longrightarrow Ra_*a^{-1}E$$

is an isomorphism.

Proof. First, let $\mathcal{I}$ be an injective abelian sheaf on $\mathcal{C}$. Then the spectral sequence of Lemma 0D7A for the sheaf $a^{-1}\mathcal{I}$ degenerates as $(a^{-1}\mathcal{I})_p = a_p^{-1}\mathcal{I}$ is injective by Lemma 0D88. Thus the complex

$$a_{0,*}a_0^{-1}\mathcal{I} \to a_{1,*}a_1^{-1}\mathcal{I} \to a_{2,*}a_2^{-1}\mathcal{I} \to \ldots$$

computes $Ra_*a^{-1}\mathcal{I}$. By Lemma 0D8F this is equal to the complex $\mathcal{H}om(s(\mathbf{Z}^{\#}_{F(K)}), \mathcal{I})$. Because K is a hypercovering, we see that $s(\mathbf{Z}^{\#}_{F(K)})$ is exact in degrees > 0 by Hypercoverings, Lemma 01GE applied to the simplicial presheaf $F(K)$. Since $\mathcal{I}$ is injective, the functor $\mathcal{H}om(-, \mathcal{I})$ is exact and we conclude that $\mathcal{H}om(s(\mathbf{Z}^{\#}_{F(K)}), \mathcal{I})$ is exact in positive degrees. We conclude that $R^p a_*a^{-1}\mathcal{I} = 0$ for $p > 0$. On the other hand, we have $\mathcal{I} = a_*a^{-1}\mathcal{I}$ by Lemma 0D8E.

Bounded case. Let $E \in D^+(\mathcal{C})$. Choose a bounded below complex $\mathcal{I}^\bullet$ of injectives representing E. By the result of the first paragraph and Leray's acyclicity lemma (Derived Categories, Lemma 015E) $Ra_*a^{-1}\mathcal{I}^\bullet$ is computed by the complex $a_*a^{-1}\mathcal{I}^\bullet = \mathcal{I}^\bullet$ and we conclude the lemma is true in this case.

Unbounded case. We urge the reader to skip this, since the argument is the same as above, except that we use explicit representation by double complexes to get around convergence issues. Let $E \in D(\mathcal{C})$. To show the map $E \to Ra_*a^{-1}E$ is an isomorphism, it suffices to show for every object U of $\mathcal{C}$ that

$$R\Gamma(U, E) = R\Gamma(U, Ra_*a^{-1}E)$$

We will compute both sides and show the map $E \to Ra_*a^{-1}E$ induces an isomorphism. Choose a K-injective complex $\mathcal{I}^\bullet$ representing E. Choose a quasi-isomorphism $a^{-1}\mathcal{I}^\bullet \to \mathcal{J}^\bullet$ for some K-injective complex $\mathcal{J}^\bullet$ on $(\mathcal{C}/K)_{total}$. We have

$$R\Gamma(U, E) = R\operatorname{Hom}(\mathbf{Z}^\#_U, E)$$

and

$$R\Gamma(U, Ra_*a^{-1}E) = R\operatorname{Hom}(\mathbf{Z}^\#_U, Ra_*a^{-1}E) = R\operatorname{Hom}(a^{-1}\mathbf{Z}^\#_U, a^{-1}E)$$

By Lemma 0D78 we have a quasi-isomorphism

$$\Big(\ldots \to g_{2!}(a_2^{-1}\mathbf{Z}^\#_U) \to g_{1!}(a_1^{-1}\mathbf{Z}^\#_U) \to g_{0!}(a_0^{-1}\mathbf{Z}^\#_U)\Big) \longrightarrow a^{-1}\mathbf{Z}^\#_U$$

Hence $R\operatorname{Hom}(a^{-1}\mathbf{Z}^\#_U, a^{-1}E)$ is equal to

$$R\Gamma((\mathcal{C}/K)_{total}, R\mathcal{H}om(\ldots \to g_{2!}(a_2^{-1}\mathbf{Z}^\#_U) \to g_{1!}(a_1^{-1}\mathbf{Z}^\#_U) \to g_{0!}(a_0^{-1}\mathbf{Z}^\#_U), \mathcal{J}^\bullet))$$

By the construction in Cohomology on Sites, Section 08J7 and since $\mathcal{J}^\bullet$ is K-injective, we see that this is represented by the complex of abelian groups with terms

$$\prod\nolimits_{p+q=n} \operatorname{Hom}(g_{p!}(a_p^{-1}\mathbf{Z}^\#_U), \mathcal{J}^q) = \prod\nolimits_{p+q=n} \operatorname{Hom}(a_p^{-1}\mathbf{Z}^\#_U, g_p^{-1}\mathcal{J}^q)$$

See Cohomology on Sites, Lemmas 0A94 and 08JA for more information. Thus we find that $R\Gamma(U, Ra_*a^{-1}E)$ is computed by the product total complex $\operatorname{Tot}_\pi(B^{\bullet,\bullet})$ with $B^{p,q} = \operatorname{Hom}(a_p^{-1}\mathbf{Z}^\#_U, g_p^{-1}\mathcal{J}^q)$. For the other side we argue similarly. First we note that

$$s(\mathbf{Z}^\#_{F(K)}) \longrightarrow \mathbf{Z}$$

is a quasi-isomorphism of complexes on $\mathcal{C}$ by Hypercoverings, Lemma 01GE. Since $\mathbf{Z}^\#_U$ is a flat sheaf of $\mathbf{Z}$-modules we see that

$$s(\mathbf{Z}^\#_{F(K)}) \otimes_{\mathbf{Z}} \mathbf{Z}^\#_U \longrightarrow \mathbf{Z}^\#_U$$

is a quasi-isomorphism. Therefore $R\operatorname{Hom}(\mathbf{Z}^\#_U, E)$ is equal to

$$R\Gamma(\mathcal{C}, R\mathcal{H}om(s(\mathbf{Z}^\#_{F(K)}) \otimes_{\mathbf{Z}} \mathbf{Z}^\#_U, \mathcal{I}^\bullet))$$

By the construction of $R\mathcal{H}om$ and since $\mathcal{I}^\bullet$ is K-injective, this is represented by the complex of abelian groups with terms

$$\prod\nolimits_{p+q=n} \operatorname{Hom}(\mathbf{Z}^\#_{K_p} \otimes_{\mathbf{Z}} \mathbf{Z}^\#_U, \mathcal{I}^q) = \prod\nolimits_{p+q=n} \operatorname{Hom}(a_p^{-1}\mathbf{Z}^\#_U, a_p^{-1}\mathcal{I}^q)$$

The equality of terms follows from the fact that $\mathbf{Z}^\#_{K_p} \otimes_{\mathbf{Z}} \mathbf{Z}^\#_U = a_{p!}a_p^{-1}\mathbf{Z}^\#_U$ by Modules on Sites, Remark 0EYY. Thus we find that $R\Gamma(U, E)$ is computed by the product total complex $\mathrm{Tot}_\pi(A^{\bullet,\bullet})$ with $A^{p,q} = \mathrm{Hom}(a_p^{-1}\mathbf{Z}^\#_U, a_p^{-1}\mathcal{I}^q)$.

Since $\mathcal{I}^\bullet$ is K-injective we see that $a_p^{-1}\mathcal{I}^\bullet$ is K-injective, see Lemma 0D88. Since $\mathcal{J}^\bullet$ is K-injective we see that $g_p^{-1}\mathcal{J}^\bullet$ is K-injective, see Lemma 09WH. Both represent the object $a_p^{-1}E$. Hence for every $p \geq 0$ the map of complexes

$$A^{p,\bullet} = \mathrm{Hom}(a_p^{-1}\mathbf{Z}^\#_U, a_p^{-1}\mathcal{I}^\bullet) \longrightarrow \mathrm{Hom}(a_p^{-1}\mathbf{Z}^\#_U, g_p^{-1}\mathcal{J}^\bullet) = B^{p,\bullet}$$

induced by g_p^{-1} applied to the given map $a^{-1}\mathcal{I}^\bullet \to \mathcal{J}^\bullet$ is a quasi-isomorphisms as these complexes both compute

$$R\,\mathrm{Hom}(a_p^{-1}\mathbf{Z}^\#_U, a_p^{-1}E)$$

By More on Algebra, Lemma 0EYX we conclude that the right vertical arrow in the commutative diagram

$$\begin{array}{ccc} R\Gamma(U, E) & \longrightarrow & \mathrm{Tot}_\pi(A^{\bullet,\bullet}) \\ \downarrow & & \downarrow \\ R\Gamma(U, Ra_*a^{-1}E) & \longrightarrow & \mathrm{Tot}_\pi(B^{\bullet,\bullet}) \end{array}$$

is a quasi-isomorphism. Since we saw above that the horizontal arrows are quasi-isomorphisms, so is the left vertical arrow. □